

- 18B.19. **Oxygen uptake by a bacterial aggregate.** Under suitable circumstances the rate of oxygen metabolism by bacterial cells is very nearly zero order with respect to oxygen concentration. We examine such a case here and focus our attention on a spherical aggregate of cells, which has a radius R . We wish to determine the total rate of oxygen uptake by the aggregate as a function of aggregate size, oxygen mass concentration ρ_0 at the aggregate surface, the metabolic activity of the cells, and the diffusional behavior of the oxygen. For simplicity we consider the aggregate to be homogeneous. We then approximate the metabolic rate by an effective volumetric reaction rate $r_{O_2} = -k_0'''$ and the diffusional behavior by Fick's law, with an effective pseudobinary diffusivity $\mathcal{D}_{O_2m}$. Because the solubility of oxygen is very low in this system, both convective oxygen transport and transient effects may be neglected.¹²

(a) Show by means of a shell mass balance that the pseudo-steady-state oxygen concentration profile is described by the differential equation

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\chi}{d\xi} \right) = N \quad (18B.19-1)$$

where $\chi = \rho_{O_2}/\rho_0$, $\xi = r/R$, and $N = k_0''' R^2 / \rho_0 \mathcal{D}_{O_2m}$.

(b) There may be an oxygen-free core in the aggregate, if N is sufficiently large, such that $\chi = 0$ for $\xi < \xi_0$. Write sufficient boundary conditions to integrate Eq. 18B.19-1 for this situation. To do this, it must be recognized that both χ and $d\chi/d\xi$ are zero at $\xi = \xi_0$. What is the physical significance of this last statement?

(c) Perform the integration of Eq. 18B.19-1 and show how ξ_0 may be determined.

(d) Sketch the total oxygen uptake rate and ξ_0 as functions of N , and discuss the possibility that no oxygen-free core exists.

Answer: (c) $\chi = 1 - \frac{N}{6}(1 - \xi^2) + \frac{N}{3}\xi_0^3 \left(\frac{1}{\xi} - 1 \right)$ for $\xi \geq \xi_0 \geq 0$, where ξ_0 is determined as a function of N from

$$\xi_0^3 - \frac{3}{2}\xi_0^2 + \left(\frac{1}{2} - \frac{3}{N} \right) = 0$$

18C.2. **Diffusion and reaction in a partially impregnated catalyst.** Consider a catalytic sphere like that in §18.7, except that the active ingredient of the catalyst is present only in the annular region between $r = \kappa R$ and $r = R$:

In region I ($0 < r < \kappa R$), $k_1' a = 0$ (18C.2-1)

In region II ($\kappa R < r < R$), $k_1' a = \text{constant} > 0$ (18C.2-2)

Such a situation may arise when the active ingredient is put on the particles after pelleting, as is done for many commercial catalysts.

(a) Integrate Eq. 18.7-6 separately for the active and inactive regions. Then apply the appropriate boundary conditions to evaluate the integration constants, and solve for the concentration profile in each region. Give qualitative sketches to illustrate the forms of the profiles.

(b) Evaluate W_{AR} , the total molar rate of conversion of A in a single particle.

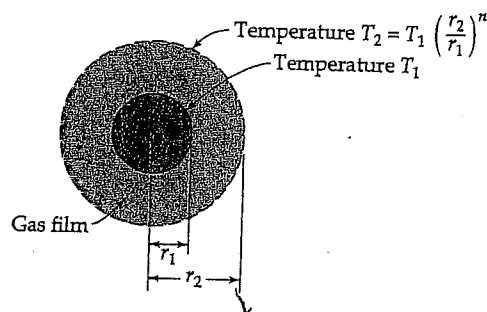
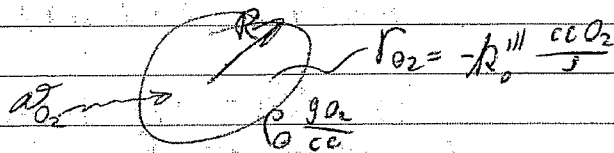


Fig. 18.2-4. Diffusion through a hypothetical spherical stagnant gas film surrounding a droplet of liquid A.

18B14 O₂ uptake



shell Balance

Out = In - consumption

$$N_{O_2 r} |_{r+\Delta r} \cdot 4\pi r^2 |_{r+\Delta r} = N_{O_2 r} |_{r-\Delta r} \cdot 4\pi r^2 |_{r-\Delta r} - R_0''' \cdot \Delta r \cdot 4\pi r^2 \Delta r$$

$$N_{O_2 r} = + a_{O_2} \frac{dR_0}{dr}$$

$$R_0''' r^2 \Delta r = \Delta (r^2 N_{O_2 r})$$

$$R_0''' r^2 |_{r-\Delta r} = \frac{d(r^2 N_{O_2 r})}{dr} \Rightarrow R_0''' r^2 = \frac{d}{dr} (r^2 N_{O_2 r})$$

$$\text{let } \xi = \frac{r}{R_0} \Rightarrow R_0''' R_0^3 \xi^2 =$$

$$R_0''' r^2 = \frac{d}{dr} (r^2 a_{O_2} \frac{dR_0}{dr}) = + a_{O_2} \frac{d}{dr} (r^2 \frac{dR_0}{dr})$$

now let $\xi = r/R_0$

$$R_0''' R_0^3 \xi^2 = + a_{O_2} \frac{d}{d\xi} (R_0^3 \xi^2 \frac{dR_0}{R_0 d\xi}) = + a_{O_2} \frac{d}{d\xi} (\xi^2 \frac{dR_0}{d\xi})$$

let $X = R_0/R_0$

$$R_0''' R_0^3 \xi^2 = + a_{O_2} R_0 \frac{d}{d\xi} (\xi^2 \frac{dX}{d\xi})$$

$$\frac{R_0''' R_0^2}{a_{O_2} R_0} = \frac{1}{\xi^2} \frac{d}{d\xi} (\xi^2 \frac{dX}{d\xi}) = N$$

for no O₂ free core.

$$X = 1 \text{ at } \xi = 1$$

$$X' = 0 \text{ at } \xi = 0$$

$$\frac{d}{d\xi} (\xi^2 \frac{dX}{d\xi}) = N \xi^2 \Rightarrow \xi^2 \frac{dX}{d\xi} = N \xi^3/3 + C_1$$

$$\text{and } X' = \frac{N}{3} \xi + C_1/\xi^2 = 0 \text{ at } \xi = 0 \Rightarrow C_1 = 0$$

$$\text{so } \frac{dX}{d\xi} = \frac{N}{3} \xi \Rightarrow X = \frac{N}{6} \xi^2 + C_2 = 1 \text{ at } \xi = 1$$

$$1 = \frac{N}{6} + C_2 \Rightarrow C_2 = 1 - N/6$$

$$X = \frac{N}{6} \xi^2 + 1 - \frac{N}{6} = 1 + \frac{N}{6} (\xi^2 - 1)$$

$$X - 1 = \frac{N}{6} (\xi^2 - 1)$$

for O_2 free core, $X=0$ and $X'=0$ at $r=r_0$

$$\begin{aligned} \text{C) } X' &= \frac{N}{3}r + C_1/r^2 = 0 \text{ at } r=r_0 \\ 0 &= \frac{N}{3}r_0 + C_1/r_0^2 \Rightarrow -C_1/r_0^2 = \frac{N}{3}r_0 \text{ and } C_1 = -\frac{N}{3}r_0^3 \\ \text{so } X' &= \frac{N}{3}r - \frac{N}{3}r_0^3/r^2 = \frac{N}{3}(r - r_0^3/r^2) \end{aligned}$$

$$\begin{aligned} X &= \frac{N}{3}\left[\frac{r^2}{2} + r_0^3 \cdot \frac{1}{r}\right] + C_2 = 0 \text{ at } r=r_0 \\ 0 &= \frac{N}{3}\left[\frac{r_0^2}{2} + r_0^2\right] + C_2 = \frac{N}{3} \cdot \frac{3}{2}r_0^2 + C_2 = \frac{N}{2}r_0^2 + C_2 \\ \Rightarrow C_2 &= -\frac{N}{2}r_0^2 \end{aligned}$$

$$\text{so } X = \frac{N}{3}\left(\frac{r^2}{2} + \frac{r_0^3}{r}\right) - \frac{N}{2}r_0^2 = \frac{N}{3} \cdot \frac{r^2}{2} + \frac{N}{3} \frac{r_0^3}{r} - \frac{N}{3} \cdot \frac{3}{2}r_0^2$$

$$X = \frac{N}{3}\left[\frac{1}{2}r^2 + r_0^2\left(\frac{r_0}{r} - \frac{3}{2}\right)\right]$$

To determine r_0 , remember $X=1$ at $r=1$

$$1 = \frac{N}{3}\left[\frac{1}{2} + r_0^2\left(r_0 - \frac{3}{2}\right)\right] = \frac{N}{3}\left[\frac{1}{2} + r_0^3 - \frac{3}{2}r_0^2\right]$$

$$\frac{3}{N} - \frac{1}{2} = r_0^3 - \frac{3}{2}r_0^2 \text{ or solve } r_0^3 - \frac{3}{2}r_0^2 + \frac{1}{2} - \frac{3}{N} = 0$$

Can the Book expression for X work for the r_0 eqn

$$\text{Book says } X = 1 - \frac{N}{6}(1-r^2) + \frac{N}{3}r_0^3\left(\frac{1}{r} - 1\right) = 1 \text{ at } r=1$$

$$1 = 1 - \frac{N}{6}(1-1^2) + \frac{N}{3}r_0^3(1-1) = 1$$

Book expression doesn't work

18.C2.

$$\nabla_A \frac{1}{r^2} \frac{d}{dr} (r^2 \frac{dC_A}{dr}) = k_1'' a C_A$$

$$\left. \begin{array}{l} R_1'' a > 0 \\ R_1'' a = 0 \end{array} \right\} \begin{array}{l} C_A = C_{AR} \text{ at } r=R \\ C_A = C_{AR} \text{ at } r=KR \\ C_A = \text{finite at } r=0 \end{array}$$

a)

$$\text{at } C_A/C_{AR} = f(r) = A$$

In the region $RR \leq r \leq R$

$$\nabla_A \frac{1}{r^2} \frac{d}{dr} (r^2 \frac{dA}{dr}) = k_1'' a A$$

$$\nabla_A \frac{1}{r^2} \frac{d}{dr} (r^2 [\frac{1}{r} f' - \frac{1}{r^2} f]) = k_1'' a \frac{f}{r}$$

$$\nabla_A \frac{1}{r^2} \frac{d}{dr} [r f' - f] = k_1'' a \frac{f}{r}$$

$$\nabla_A \frac{1}{r^2} [r f'' + f' - f'] = k_1'' a \frac{f}{r}$$

$$\nabla_A \frac{1}{r^2} f'' = k_1'' a \frac{f}{r}$$

$$\nabla_A f'' = k_1'' a f \Rightarrow f'' - \left(\frac{R_1'' a}{\nabla_A} \right) f = 0$$

$$f = C_1 \sinh br + C_2 \cosh br$$

$$f'' = b^2 f$$

$$b^2 f - \left(\frac{R_1'' a}{\nabla_A} \right) f = 0 \Rightarrow b = \sqrt{\frac{R_1'' a}{\nabla_A}}$$

$$\frac{C_A}{C_{AR}} = \frac{1}{r} [C_1 \sinh br + C_2 \cosh br]$$

$$1 = R [C_1 \sinh bR + C_2 \cosh bR] \Rightarrow C_2 = \frac{R - C_1 \sinh bR}{\cosh bR}$$

$$\frac{C_{AK}}{C_{AR}} = \frac{1}{KR} [C_1 \sinh bKR + C_2 \cosh bKR] \Rightarrow C_1 = f(C_{AK}, KR)$$

In the region $0 \leq r \leq RR$

$$\nabla_A \frac{1}{r^2} \frac{d}{dr} (r^2 \frac{dC_A}{dr}) = 0 = \nabla_A \frac{1}{r^2} \frac{d}{dr} (r^2 \frac{dC_3}{dr})$$

$$\frac{d}{dr} (r^2 \frac{dC_3}{dr}) = 0$$

$$\frac{dC_3}{dr} = C_3 / r^2$$

$$\Rightarrow C_A = -\frac{C_3}{r} + C_4 \quad \text{and } C_A \text{ is finite at } r=0 \Rightarrow C_3=0$$

$$C_A = C_4 = C_{AR} \text{ since } C_A = C_{AR} \text{ at } r=RR$$

As ^{2nd} matching condition at $r=KR$, we could require dC_A/dr to match coming from both directions, i.e. $\frac{dC_A}{dr} \Big|_{r \rightarrow KR^-} = \frac{dC_A}{dr} \Big|_{r \rightarrow KR^+}$.

And $\frac{dC_A}{dr} \Big|_{r \rightarrow KR^-} = 0$ since $C_A = C_{AR}$ in $0 \leq r \leq KR$.

$$\frac{C_A}{CAR} = \frac{1}{r} [C_1 \sinh br + C_2 \cosh br]$$

$$C_A' = CAR \left\{ \frac{b}{r} [C_1 \cosh br + C_2 \sinh br] - \frac{1}{r^2} [C_1 \sinh br + C_2 \cosh br] \right\}$$

and $C_A = 0$ at $r = KR$

$$0 = b [C_1 \cosh bKR + C_2 \sinh bKR] - \frac{1}{KR} [C_1 \sinh bKR + C_2 \cosh bKR]$$

Since C_1 & C_2 have CAR in them, we can solve for CAR

To do this, remember

$$\frac{C_A}{CAR} = \frac{1}{KR} [C_1 \sinh bKR + \left(\frac{KR - C_1 \sinh bR}{\cosh bR} \right) \cosh bKR]$$

$$KR \frac{C_A}{CAR} = C_1 [\sinh bKR - \tanh bR \cosh bKR] + R \frac{\cosh bKR}{\cosh bR}$$

$$\frac{KR \frac{C_A}{CAR} - R \frac{\cosh bKR}{\cosh bR}}{\sinh bKR - \tanh bR \cosh bKR} = C_1 \quad \& \quad C_2 = \frac{KR - C_1 \sinh bR}{\cosh bR}$$

$$C_5 C_A - C_6 = C_1 \quad \& \quad C_2 = C_7 - \frac{C_8}{\sinh bR} (C_5 C_A - C_6)$$

so,

$$0 = b [\cosh bKR (C_5 C_A - C_6) + \sinh bKR (C_7 + C_8 C_6 - C_8 C_5 C_A)]$$

$$- \frac{1}{KR} [\sinh bKR (C_5 C_A - C_6) + \cosh bKR (C_7 + C_8 C_6 - C_8 C_5 C_A)]$$

$$0 = b \cosh bKR \cdot C_5 C_A - b \sinh bKR \cdot C_6 C_5 C_A - \frac{\sinh bKR}{KR} \cdot C_5 C_A + \frac{\cosh bKR}{KR} \cdot C_6 C_5 C_A$$

$$- b \cosh bKR \cdot C_6 + b \sinh bKR (C_7 + C_8 C_6) + \frac{\sinh bKR}{KR} \cdot C_6 - \frac{\cosh bKR}{KR} (C_7 + C_8 C_6)$$

$$0 = CAR \left\{ C_5 \cosh bKR \left(b + \frac{C_8}{KR} \right) - C_5 \sinh bKR \left(b C_6 + \frac{1}{KR} \right) \right\}$$

$$+ \left\{ - \cosh bKR \left(b \cdot C_6 + \frac{C_7 + C_8 C_6}{KR} \right) + \sinh bKR \left(b (C_7 + C_8 C_6) + \frac{C_6}{KR} \right) \right\}$$

$$0 = \alpha CAR + \beta \Rightarrow CAR = -\frac{\beta}{\alpha}$$

$$b) W_{AR} = 4\pi R^2 N_{AR} \Big|_{r=R} = -4\pi R^2 d_A \frac{dC_A}{dr} \Big|_{r=R}$$

$$\frac{C_A}{C_{AR}} = \frac{1}{r} [C_1 \sinh br + C_2 \cosh br]$$

$$\begin{aligned} \frac{d}{dr} \left(\frac{C_A}{C_{AR}} \right) &= \frac{b}{r} [C_1 \cosh br + C_2 \sinh br] - \frac{1}{r^2} [C_1 \sinh br + C_2 \cosh br] \\ &= \sinh br \left[\frac{C_2 b}{r} - \frac{C_1}{r^2} \right] + \cosh br \left[\frac{C_1 b}{r} - \frac{C_2}{r^2} \right] \end{aligned}$$

$$\text{at } r=R, \frac{dC_A}{dr} = \frac{C_{AR}}{R} \left\{ \sinh bR \left(C_2 b - \frac{C_1}{R} \right) + \cosh bR \left(C_1 b - \frac{C_2}{R} \right) \right\}$$

$$W_{AR} = 4\pi R C_{AR} d_A \left\{ \sinh bR \left(C_2 b - \frac{C_1}{R} \right) + \cosh bR \left(C_1 b - \frac{C_2}{R} \right) \right\}$$